

Gradient-Aligned Iterative Fine-Tuning for Adaptive LLM Post-Training

Zhan Shi

Amazon / Canada

shizhan@amazon.com

Abstract

Post-training has become the primary mechanism for grounding large language models (LLMs) to application-specific domains, enabling them to acquire new knowledge, behaviors, and policies beyond pre-training. Supervised fine-tuning (SFT) and preference optimization are the dominant paradigms for this adaptation, but are typically applied sequentially, requiring careful tuning of training schedules, regularization, and other hyperparameters.

We propose *Gradient-Aligned Iterative Fine-Tuning (GAIF)*, an adaptive post-training framework that combines three complementary components: a unified SFT+DPO objective optimized simultaneously; a prospective adaptive weighting mechanism that selects task weights based on next-step gradient geometry; and an iterative outer loop that regenerates preference pairs from the current model’s failures, keeping the supervision signal synchronized with the model’s evolving capabilities while adapting training toward arbitrary evaluation metrics, including non-differentiable ones.

We evaluate GAIF on a production-scale generative movie search assistant, where a 7B LLM is adapted to the entertainment domain while remaining faithful to a fixed streaming catalog. The 7B model trained with GAIF outperforms standard SFT, sequential SFT+DPO, and a ~ 90 B reference model in both ranking quality (NDCG) and output validity. Experiments on GSM8K mathematical reasoning further show the algorithm’s effectiveness extends beyond grounded retrieval.

1 Introduction

Large language models (LLMs) are increasingly adapted to domain-specific applications through post-training techniques such as supervised fine-tuning (SFT) and preference optimization (Lai et al., 2025; Rafailov et al., 2023). Rather than relying solely on the general capabilities acquired dur-

ing pre-training, these methods specialize a foundation model to produce outputs that are faithful to a downstream domain, policy, or evaluation criterion (Ye et al., 2024; Carta et al., 2023). Such adaptation has become a fundamental component of modern LLM systems (Lai et al., 2025), enabling applications ranging from domain-specific assistants and retrieval systems to recommendation, tool use, and reasoning.

Despite their widespread adoption, existing post-training methods face several practical challenges. First, high-quality supervision is expensive to obtain, particularly human-curated preference data, making it important to maximize the value of limited annotations. Second, post-training is inherently iterative (Ouyang et al., 2022): practitioners commonly optimize supervised and preference objectives in separate stages, yet improvements in one stage do not necessarily translate to better performance in the next, often requiring repeated retraining and extensive hyperparameter tuning. Finally, post-training targets are rarely static. As models improve, they exhibit different failure modes, while application-specific evaluation metrics and judge rubrics frequently evolve alongside production systems. Consequently, supervision that is effective early in training or under one evaluation protocol may become less informative later.

These observations suggest that post-training should itself be adaptive. Rather than optimizing against a fixed supervision signal using a fixed training procedure, we ask whether the optimization process can continuously adapt to both the evolving model and the application’s notion of correctness throughout training.

To this end, we propose Gradient-Aligned Iterative Fine-Tuning (GAIF), an adaptive post-training framework comprising three complementary components: a unified SFT+DPO objective optimized jointly; a prospective adaptive weighting mechanism that selects task weights by maximizing next-

step gradient alignment; and an iterative outer loop that regenerates preference pairs from the current model’s failures using an application-defined evaluation function. Together, these components enable GAIF to adapt its optimization strategy, supervision signal, and optimization target throughout training without requiring additional supervision or computationally expensive reinforcement learning.

Our contributions are:

1. **GAIF**, a fine-tuning algorithm that automatically adapts to data, metric and gradient geometry. It makes efficient use of positive SFT labels and existing evaluator while requires minimal extra tuning effort than standard SFT.
2. **Empirical validation** on a production movie search task and the GSM8K benchmark, demonstrating consistent improvements over SFT and sequential SFT+DPO baselines across both task performance and output validity metrics.

2 Related Work

Grounded LLM Adaptation. Recent work has explored grounding LLMs to external knowledge, environments, and application-specific constraints through retrieval augmentation, tool use, and interaction with external systems. Learning-based approaches such as AGREE adapt LLMs through post-training to generate responses grounded in retrieved evidence and accompanied by faithful citations (Ye et al., 2024). Other work grounds language models through continual interaction with external environments using online reinforcement learning (Carta et al., 2023). These approaches primarily focus on designing mechanisms that connect language models to external sources of information or feedback. Our work is complementary: rather than introducing a new grounding mechanism, we study how the post-training process itself can adapt to evolving application-specific grounding objectives by jointly evolving the optimization strategy and supervision signal.

LLM post-training and preference optimization. Supervised fine-tuning and preference optimization are central tools for adapting language models to downstream tasks. RLHF improves instruction following by training from human feedback (Ouyang et al., 2022), while DPO replaces reward-model training and reinforcement learning with a direct

classification-style preference objective (Rafailov et al., 2023). Subsequent methods simplify or modify preference optimization further; for example, ORPO integrates an odds-ratio preference penalty into supervised learning without a separate reference model (Hong et al., 2024), while SimPO uses a reference-free reward based on average sequence log-probability (Meng et al., 2024). These methods improve preference alignment, but are usually applied as separate post-training stages or with fixed objective choices. GAIF instead studies how supervised and preference objectives can be optimized jointly with an adaptive weighting rule that responds to the current model and optimization trajectory.

Iterative preference generation and self-improvement. A separate line of work studies how preference data should evolve as the policy changes. Preference fine-tuning can suffer when rejected samples are stale or too far from the current model distribution (Tajwar et al., 2024). Iterative methods address this issue by regenerating preference data from the current model (Xiong et al., 2024; Yuan et al., 2024), a principle also reflected in inference-time self-improvement procedures (Madaan et al., 2023; Shinn et al., 2023). In recommendation, SPRec similarly constructs preference pairs from current model outputs for self-play recommendation (Gao et al., 2025). GAIF shares the principle of keeping preference data synchronized with the current model, but couples this outer-loop regeneration with an inner-loop adaptive optimization rule. The evaluator is also task-defined: preference pairs are constructed only when the current model underperforms the positive label by a margin, so the same scoring function used for evaluation drives the training signal.

Adaptive multi-task optimization. Jointly optimizing SFT and DPO can be viewed as a two-objective optimization problem. Existing multi-task learning methods mitigate gradient imbalance or conflict using the geometry observed at the current iterate. GradNorm balances training rates through gradient magnitudes (Chen et al., 2018); MGDA searches for a minimum-norm element in the convex hull of task gradients (Désidéri, 2012); PCGrad projects away conflicting gradient components (Yu et al., 2020); CAGrad optimizes conflict-averse updates (Liu et al., 2021); and CoGrad promotes knowledge transfer by coordinating task gradients (Yang et al., 2023). These methods are adap-

tive, but they are primarily current-state methods: the update is determined from the gradient geometry before the step is taken. GAIF instead uses a prospective objective, selecting task weights by maximizing the predicted alignment of task gradients at the next iterate. Closely related look-ahead approaches such as L-DRO optimize future loss-based objectives (Michel et al., 2021); GAIF differs by making future gradient alignment itself the objective and connecting it to common descent.

3 Method

3.1 Problem Setup

We consider the general problem of post-training a language model π_θ for a downstream task. Given an input x , the model generates an output $y = \pi_\theta(x)$. The task may correspond to retrieval, recommendation, reasoning, or any other problem for which candidate outputs can be evaluated.

GAIF assumes access to two forms of supervision. The first is a set of positive training examples

$$\mathcal{D} = \{(x, y^+)\},$$

where y^+ denotes a desirable response for input x . Depending on the application, these labels may originate from human annotation, implicit or explicit user feedback, or outputs from a stronger reference model. GAIF is agnostic to the provenance of these labels and only requires positive demonstrations suitable for supervised fine-tuning.

The second ingredient is an evaluation function

$$s(x, y) \in \mathbb{R},$$

that measures the quality of a generated response. The evaluation criterion is task dependent and may be implemented using exact-match metrics, ranking metrics such as NDCG, F1 scores, program execution, LLM-as-a-Judge (Zheng et al., 2023), or other application-specific evaluators. Unlike the supervised labels, which specify desirable demonstrations, the evaluation function defines the optimization target that guides iterative preference generation. Consequently, the quality of the evaluator is critical: improvements under the evaluation function should correlate with improvements in the desired downstream behavior. In practical applications, this may correspond to agreement with human preferences, online production metrics, or other task-specific notions of success.

Given an input x , GAIF generates one or more candidate outputs from the current model and uses

the evaluation function to identify preferred and rejected responses. These preference pairs are then used for preference optimization, while the positive labels continue to supervise the SFT objective. The goal is to optimize the expected evaluation score

$$\max_{\theta} \mathbb{E}_x [s(x, \pi_\theta(x))],$$

while preserving faithful and task-appropriate generation behavior.

3.2 Algorithm Overview

GAIF trains the model through an inner-outer loop structure motivated by two complementary objectives. The inner loop optimizes a combined loss that simultaneously teaches the model what good outputs look like and steers it away from pathological behaviors. The outer loop periodically evaluates the current model, identifies its failure modes, and regenerates preference pairs from those failures before the next inner loop begins. This design reflects a simple intuition: a model that has just learned to avoid one set of failure modes will develop new, subtler ones as it improves, and the training signal should track those evolving failures rather than remaining anchored to the errors of the initialization (Tajwar et al., 2024).

Concretely, at outer loop iteration i , the current model π_{θ_i} generates candidate outputs for a sample of queries; those outputs are scored by the evaluation function s ; query-output pairs where the model underperforms relative to y^+ by a margin δ are selected as rejected samples y^- ; and the resulting preference dataset $\mathcal{D}_i = \{(x, y^+, y^-)\}$ is used to train the inner loop. Because y^- is always generated by $\pi_{\theta_{i-1}}$, the preference pairs reflect the model’s current failure modes rather than those at initialization, resembling on-policy training and iterative self-improvement procedures (Ross et al., 2011; Madaan et al., 2023; Shinn et al., 2023). A further consequence of the outer loop is *metric alignment*: because y^- selection is driven by the scoring function s , which can incorporate any downstream pipeline including non-differentiable steps such as entity resolution, the training signal reflects end-to-end task performance rather than a proxy objective.

The inner loop optimizes a unified SFT+DPO loss with adaptive weighting, described in the following subsections, producing updated parameters θ_{i+1} . These parameters serve as both the new model and the reference policy for the next outer

loop iteration. The full procedure is summarized in Algorithm 1.

Algorithm 1 GAIF

- 1: Initialize θ_0 from a pretrained LLM
 - 2: **for** outer iteration $i = 1, 2, \dots$ **do**
 - 3: Sample queries $\{x_j\}$ from the query distribution
 - 4: Generate outputs $\hat{y}_j = \pi_{\theta_{i-1}}(x_j)$
 - 5: Score outputs: $s_j = s(x_j, \hat{y}_j)$
 - 6: Select failures: $y_j^- \leftarrow \hat{y}_j$ where $s_j < s(x_j, y_j^+) - \delta$
 - 7: Construct $\mathcal{D}_i = \{(x_j, y_j^+, y_j^-)\}$
 - 8: Run inner loop for T steps on $\mathcal{D}_i$ with adaptive weights, using $\pi_{\theta_{i-1}}$ as reference policy
 - 9: Set $\theta_i \leftarrow$ trained parameters
 - 10: **end for**
-

3.3 Inner Loop: Unified SFT+DPO Loss

At each outer loop iteration i , the inner loop optimizes a combined objective over the preference dataset $\mathcal{D}_i$. Rather than treating SFT and DPO as sequential stages, GAIF trains both simultaneously against the same labels:

$$\mathcal{L}(\theta) = w_{\text{SFT}} \mathcal{L}_{\text{SFT}}(\theta) + w_{\text{DPO}} \mathcal{L}_{\text{DPO}}(\theta) + w_{\text{KL}} \mathcal{L}_{\text{KL}}(\theta, \theta_i). \quad (1)$$

where $\mathcal{L}_{\text{SFT}}(\theta) = -\log \pi_{\theta}(y^+|x)$ is the standard SFT loss on positive labels, and $\mathcal{L}_{\text{DPO}}$ is the DPO loss on preference pairs (y^+, y^-) . The explicit KL term provides a stable regularization floor that is decoupled from the adaptive balance between SFT and DPO:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \left[\log \frac{\pi_{\theta}(y^+|x)}{\pi_{\theta_i}(y^+|x)} - \log \frac{\pi_{\theta}(y^-|x)}{\pi_{\theta_i}(y^-|x)} \right] \right). \quad (2)$$

$\mathcal{L}_{\text{SFT}}$ improves the model by directly maximizing the likelihood of y^+ while $\mathcal{L}_{\text{DPO}}$ by widening the margin between y^+ and y^- . The cooperative structure between the two is important. Figure 1 illustrates what happens when DPO is dominating: after an initial improvement, the model enters a reward hacking (Amodei et al., 2016; Skalse et al., 2022) regime where it widens the margin by simultaneously reducing the logp of both y^+ and y^- . Task performance collapses sharply at this point.

3.4 Adaptive Weighting via Gradient Alignment

While w_{SFT} and w_{DPO} can be treated as fixed hyperparameters, doing so requires careful tuning and cannot adapt to the shifting gradient geometry across training. Therefore, we set $w_{\text{DPO}} = 1 - w_{\text{SFT}}$ and let $w_{\text{SFT}} \in [0, 1]$ vary adaptively¹. This eliminates the need to tune task weights and relaxes sensitivity to other hyperparameters such as the DPO β , as the algorithm compensates automatically. Concretely, at each step we select w_{SFT} to maximize gradient alignment at the *next* parameter update:

$$w_{\text{SFT}} \leftarrow \arg \max_{w_{\text{SFT}} + w_{\text{DPO}} = 1} \langle g_{\text{SFT}}(\theta - \eta g), g_{\text{DPO}}(\theta - \eta g) \rangle \quad (3)$$

where η is the learning rate, $g := \nabla_{\theta} \mathcal{L}$ is a function of w_{SFT} and $g_t := \nabla_{\theta} \mathcal{L}_t, t \in \{\text{SFT}, \text{DPO}\}$. The *prospective* nature of this update rule is critical as reactive methods that fix gradient updates after conflict occurs are insufficient under our setting as demonstrated in Section 5.1, we will discuss more about the motivation in the next section.

In order to make the above objective tractable, we apply a first-order Taylor expansion to the task gradients and approximate each task Hessian with the rank-1 Gauss-Newton estimate $H_t \approx \hat{g}_t \hat{g}_t^{\top}$; we also drop the KL divergence term for weight selection. This gives:

$$g_t(\theta - \eta g) \approx \hat{g}_t(\theta) - \eta(\hat{g}_t^{\top} \hat{g}) \hat{g}_t(\theta), \quad (4)$$

where $\hat{g}, \hat{g}_t$ indicate per-batch gradients. This yields a closed-form quadratic in w_{SFT} , with a unique stationary point at:

$$w^* = \frac{\eta(ab - 2bc + c^2) + (b - a)}{2\eta(a - c)(b - c)}, \quad (5)$$

$$a := \|\hat{g}_S\|^2, \quad b := \|\hat{g}_D\|^2, \quad c := \langle \hat{g}_S, \hat{g}_D \rangle.$$

whenever $(a - c)(b - c) \neq 0$. Consequently, the optimal weight is selected among $\{w^*, 0, 1\}$, whichever yields higher alignment.

A more expressive alternative uses the diagonal empirical Fisher $H \approx \text{diag} \left(\frac{1}{N} \sum_{i=1}^N g_i \odot g_i \right)$ (Kirkpatrick et al., 2017) to estimate the Hessian, which better captures gradient curvature but requires per-example gradients, incurs higher memory cost, and requires a grid search over w_{SFT} rather

¹Optimizing against a moving objective means convergence is to a neighborhood of the regularized Pareto front rather than the front itself, with the neighborhood radius depending on the rate of weight change

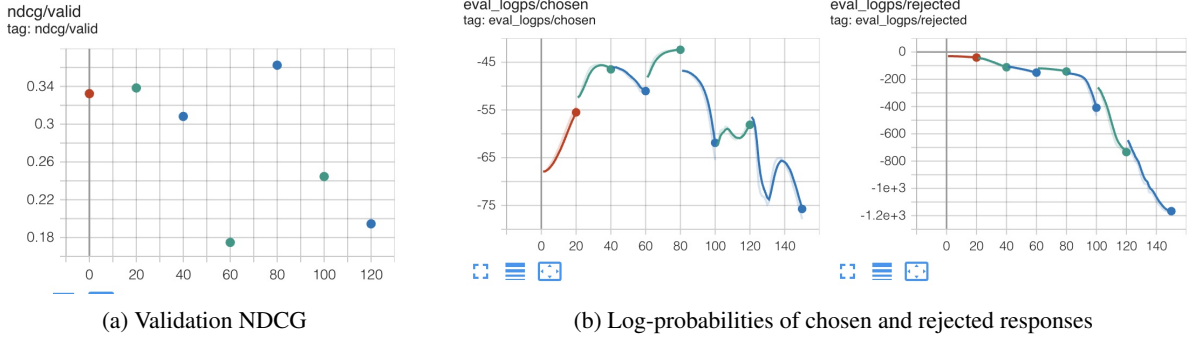


Figure 1: Training dynamics when DPO dominates the training on the movie search task ($w_{DPO} = 0.9$, $w_{SFT} = 0.1$). NDCG on the validation set collapses after approximately step 80 (left). The simultaneous collapse of log-probabilities for both chosen y^+ and rejected y^- responses (right) is the signature of reward hacking: rather than increasing the likelihood of y^+ , the model widens the margin by driving both toward $-\infty$.

than a closed-form solution. The tighter approximation extends the regime of maintained alignment, as confirmed empirically on GSM8K in Section 5.

4 Motivation of the Adaptive Weighting Scheme

Section 3.4 introduced next-step gradient alignment as the weight-selection objective. We now justify this choice: we show why reactive, current-iterate methods fail in the SFT+DPO setting, and why maximizing next-step alignment is the principled alternative.

4.1 Limitations of Reactive Methods

Existing multi-task gradient methods operate reactively: they observe the gradient geometry at the current iterate θ_t and modify either the gradients or the task weights to resolve conflicts that have already arisen. GradNorm (Chen et al., 2018) balances gradient norms across tasks, ensuring no single task’s gradient dominates in magnitude. PCGrad (Yu et al., 2020) projects conflicting gradients onto each other’s normal planes, removing the antagonistic component. CoGrad (Yang et al., 2023) quantifies the loss reduction one task’s gradient induces on another and adds an auxiliary gradient to maximize it. MGDA (Désidéri, 2012) finds the minimum-norm element in the convex hull of task gradients, yielding the steepest common descent direction.

These methods share a common premise: that the current gradient information is informative about what the update should be. This premise holds when the two task losses have comparable structure, such as similar curvature, scale, or loss progression. When the losses are structurally dis-

similar, the current gradient geometry can be misleading. The SFT+DPO setting is a severe case. The SFT loss is a cross-entropy: its gradients are relatively small and low-variance, yet it progresses steadily and can be driven close to zero. The DPO loss, a log-sigmoid of a log-ratio margin, behaves oppositely — its gradients are larger and noisier, but its loss decreases orders of magnitude more slowly and plateaus well above zero. Gradient norm and actual progress thus point in opposite directions. Reactive methods, which read only the current geometry, are misled by this mismatch. GradNorm, for instance, balances training rates across tasks; because DPO’s loss progresses so slowly, it assigns DPO disproportionately high weight (>0.99 in our runs). This is precisely the wrong response: it drives the model into the reward-hacking regime shown in Figure 1. PCGrad, applied in combination with GradNorm, cannot correct this because it intervenes post-hoc: once GradNorm has already driven the model into a bad region of parameter space, projecting away the conflicting gradient component is insufficient to recover (Table 1).

4.2 The Prospective View

The failure of reactive methods motivates a different question: rather than asking how to combine the gradients at θ_t , ask which θ_{t+1} to move to. The weight w_t determines $\theta_{t+1}(w_t) = \theta_t - \eta \nabla F_{w_t}(\theta_t)$, so selecting w_t is equivalent to selecting the next iterate. The question becomes: what makes one next iterate better than another?

A key requirement is that the update should make progress on both tasks simultaneously. This depends not only on the magnitudes of the task

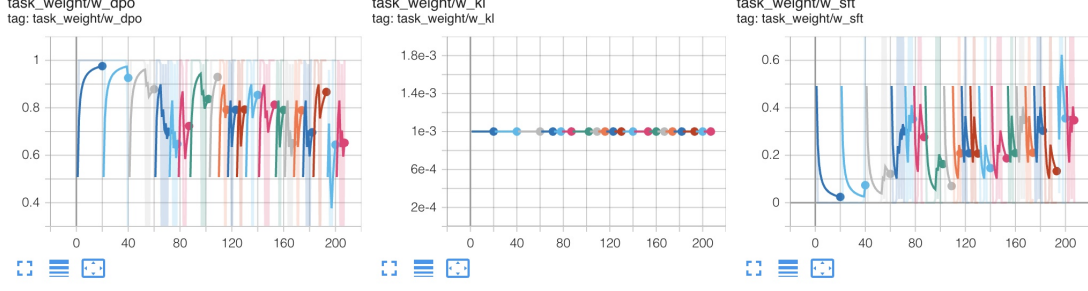


Figure 2: Adaptive task weights across inner loop training steps over multiple outer loop iterations (each color represents one outer loop iteration). w_{DPO} (left) starts dominant but trends downward in later outer loop iterations; w_{KL} (center) is held constant at 10^{-3} .

gradients at θ_{t+1} , but on how well their directions agree. If the gradients are aligned, then any convex combination of them yields a descent direction for both losses; if they are opposed, improving one task may necessarily worsen the other. A simple measure of such alignment is their inner product $\langle g_{\text{SFT}}(\theta_{t+1}), g_{\text{DPO}}(\theta_{t+1}) \rangle$. It directly measures how compatible the two optimization directions are. Maximizing this quantity therefore naturally encourages future updates to remain cooperative. The next section formalizes this intuition by showing that next-step gradient alignment is not merely a heuristic measure of cooperation, but characterizes when universal common descent is achievable.

4.3 Why Gradient Alignment

At each iterate θ_t , denote:

$$g_S^t = \nabla \mathcal{L}_S(\theta_t), \quad g_D^t = \nabla \mathcal{L}_D(\theta_t), \\ a_t = \|g_S^t\|^2, \quad b_t = \|g_D^t\|^2, \quad c_t = \langle g_S^t, g_D^t \rangle.$$

Definition 1 (Universal common descent). *We say universal common descent holds at θ_t if, for every mixing weight $w' \in [0, 1]$, the combined task gradient $w'g_S^t + (1 - w')g_D^t$ is a descent direction for both losses.*

Proposition 1 (Alignment is necessary and sufficient for universal common descent). *Universal common descent holds at θ_t if and only if $c_t = \langle g_S^t, g_D^t \rangle > 0$ (assuming $g_S^t \neq 0$ and $g_D^t \neq 0$).*

Proof sketch. For any $w' \in [0, 1]$, $\langle w'g_S^t + (1 - w')g_D^t, g_S^t \rangle = w'a_t + (1 - w')c_t$ and symmetrically $w'c_t + (1 - w')b_t$ for g_D^t . With $a_t, b_t \geq 0$, both are positive for all w' iff $c_t > 0$ (necessity follows from the endpoints $w' = 1$ and $w' = 0$).

This proposition characterizes the cooperative regime at a fixed iterate. We now lift this criterion to the next iterate induced by the current weight

choice. Denote the true next-step alignment as a function of the current weight:

$$c_{t+1}(w) := \langle g_S(\theta_{t+1}(w)), g_D(\theta_{t+1}(w)) \rangle$$

where $\theta_{t+1}(w) = \theta_t - \eta \nabla F_w(\theta_t)$. By Proposition 1, $c_{t+1}(w) > 0$ if and only if universal common descent holds at $\theta_{t+1}(w)$. Therefore, GAIF (the ideal objective without Hessian approximation) ensures the common descent at next step whenever it is possible. When universal common descent is not achievable, a principled objective is to maximize the measure of next-step mixing weights $w' \in [0, 1]$ for which common descent holds. Specifically, the feasible set of next-step mixtures is

$$\left(\frac{-c_{t+1}}{a_{t+1} - c_{t+1}}, \frac{b_{t+1}}{b_{t+1} - c_{t+1}} \right) \cap [0, 1].$$

whose measure $\mu(w)$ satisfies, When $c_{t+1} < 0$,

$$\mu(w) = \frac{a_{t+1}b_{t+1} - c_{t+1}^2}{(a_{t+1} - c_{t+1})(b_{t+1} - c_{t+1})}.$$

In particular, for fixed a_{t+1} and b_{t+1} , $\mu(w)$ is strictly increasing in c_{t+1} . Thus alignment is the exact control variable for the feasible-set size whenever the task-gradient norms are held fixed. More generally, if variation in c_{t+1} (in relations to w) dominates variation in a_{t+1} and b_{t+1} , then maximizing c_{t+1} is also the correct local objective for enlarging the feasible region. Empirically, in the SFT-DPO setting, gradient norms vary substantially less across candidate next iterates than gradient alignment, which motivates using next-step alignment as the optimization target.

5 Experiments

We evaluate GAIF on two tasks: a production movie search assistant and the GSM8K mathematical reasoning benchmark. The former demonstrates

practical effectiveness in a real-world retrieval setting; the latter validates that the algorithm generalizes beyond retrieval.

5.1 Movie Search Assistant

Task. The goal is to fine-tune a language model to serve as a movie search assistant: given a natural language query such as “feel-good movies to watch on a rainy day”, the model generates up to six movie or TV titles as a ranked list. A production semantic search system is currently in use, and a small-scale online test has shown that a LLM (Nova Pro, $\sim 90\text{B}$ parameters²) can outperform it. However, Nova Pro cannot be deployed at full traffic scale due to latency and throughput constraints. The objective is therefore to develop a fine-tuned LLM that matches Nova Pro’s retrieval quality while remaining low-latency and can be self-hosted by the platform.

Setup. We finetune Mistral-7B (Jiang et al., 2023) using LoRA with $\alpha = \text{rank} = 16$. Training queries are sampled from production search traffic from a large video-streaming platform, with short or malformed queries filtered out (e.g., single characters or incomplete phrases). This yields approximately 26K queries with over 95% traffic-weighted coverage. Positive labels y^+ are generated by Claude 3.7 Sonnet, which serves both as the teacher model and judge.

For GAIF, at each outer loop iteration, 2,048 queries are randomly sampled from the training set. The current model and the teacher model each generate a ranked list, which goes through entity resolution to map to catalog items (via RapidFuzz). They are then scored by an LLM judge using a detailed relevance rubric³. Pairs where the NDCG scores differ by more than 0.1 are retained as preference pairs, with the higher-scoring output assigned as y^+ . The inner loop trains on the sampled pairs for up to 10 epochs with an 80/20 train-validation split, using early stopping with patience 3. The adaptive weights are computed using rank-1 GN approximation as described in Equation 5. Learning rate ($2\text{e-}5$) and batch size (512) are selected via grid search; KL regularization is fixed at 10^{-3} . For the fixed-weight ablation, we searched

$w_{\text{SFT}} \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$ under the constraint $w_{\text{DPO}} + w_{\text{SFT}} = 1$, with equal weighting performing best. We run 20 outer loop iterations: data selection and inner-loop early stopping together stabilize training and performance plateaus after approximately 10-15 outer loop iterations, with further iterations yielding diminishing returns.

Evaluation. We evaluate on a human-curated set of 2,000 queries, disjoint from training and balanced across three types: title queries (e.g., a specific title), attribute-based queries (e.g., “Tom Cruise”), and complex intent queries (e.g., “relaxing movie for weekends”). The primary metric is NDCG@6 computed after fuzzy entity resolution against the catalog, with relevance scores provided by the aforementioned LLM judge. We also report the average number of unresolved items per query (titles not mappable to the catalog, e.g. hallucinations) and the average number of valid items per query (affected by output length, duplicates, and hallucinations), providing a more complete picture of output quality beyond ranking.

Candidates. We compare GAIF against: (1) Base Mistral-7B; (2) SFT only; (3) sequential SFT-then-DPO; (4) GAIF but instead of using the proposed adaptive weighting, we use GradNorm and PCGrad applied to the combined SFT+DPO loss; (5) GAIF but with fixed task weights throughout training, where $w_{\text{SFT}} = w_{\text{DPO}}$; and (6) Nova Pro, the independent large model baseline.

Table 1: NDCG@6, average unresolved titles, and average valid titles per query on the movie search assistant evaluation set. Confidence intervals obtained through bootstrapping.

Method	NDCG@6, CI95	Unres.	Valid
Base Mistral-7B	(0.4545, 0.4633)	2.234	4.924
SFT only	(0.4746, 0.4837)	2.703	3.167
Sequential SFT + DPO	(0.4539, 0.4634)	2.256	3.705
GradNorm + PC-Grad	(0.2701, 0.2720) [†]	–	–
GAIF, fixed weights	(0.4761, 0.4878)	1.931	3.833
Nova Pro	(0.4729, 0.4822)	2.008	3.949
GAIF (ours)	(0.4891, 0.4986)	1.7185	4.290

[†] DPO dominates the training and training collapsed similar to Figure 1. One notable failure mode is that the model defaulted to generating only “assistant [EOS]” for many inputs.

Results. SFT alone improves NDCG substantially but at the cost of output validity. Its high unresolved title count and low valid title count would be unacceptable in a production system where ev-

²Amazon has not disclosed the parameter count. This figure is a community estimate (Thompson, 2024).

³The rubric, as well as the relevance-score post-processing, is calibrated against online search experiments to maximize correlation with online engagement metrics. This LLM-judge is developed independent of this experiment.

ery hallucinated or missing result directly degrades user experience. Sequential SFT-then-DPO recovers output validity but fails to improve NDCG over SFT or even the base model, demonstrating the structural tension between the two objectives. GradNorm, which assigns disproportionately high weight (>0.99) to DPO due to its much slower loss progression, collapses during training even when combined with PCGrad, demonstrating that reactive gradient correction cannot recover the training once task gradients begin to conflict. This further motivates prospective task weighting, which maximizes the likelihood of gradient alignment throughout training. GAIF with fixed task weights already outperforms all baselines on both NDCG and output validity, and full GAIF improves further. As shown in Figure 2, GAIF assigns high initial weight to DPO but dynamically balances with SFT as the gradient geometry evolves. Notably, despite aggressive DPO weighting, GAIF never collapsed across any of our hyperparameter search runs. This contrasts sharply with pure DPO training, which collapses readily under similar conditions due to reward hacking.

5.2 GSM8K

To evaluate whether GAIF generalizes beyond recommendation, we apply it to GSM8K (Cobbe et al., 2021) using Qwen2.5-1.5B-Instruct (Qwen et al., 2024) as the base model. The evaluation and training pair generation require no external evaluator: positive labels are ground-truth answers with CoT, and evaluation is exact match after parsing. Preference pairs are constructed from the model’s incorrect output versus ground-truth solutions. This task requires the model to follow a specific output format in addition to producing correct answers. We use the following zero-shot system prompt:

You are a helpful assistant. Please think through the problem step by step before answering. Then provide only the final answer after the delimiter “####”.

Output length is restricted to 200 tokens. Answers are extracted from the number after the last occurrence of ####. This setup requires the model to simultaneously reason correctly, remain concise, and follow the output format. The base Qwen model struggles significantly with this joint requirement: despite having some mathematical capability, its verbosity and poor instruction following cause it to rarely produce a parseable answer within the token

budget, resulting in a near-zero success rate in this task.

We compare SFT against two variants of GAIF that differ in the Hessian approximation used for adaptive weighting. The first uses the rank-1 Gauss-Newton estimate, which admits a closed-form solution for w_{SFT}^* . The second uses the diagonal empirical Fisher $H \approx \text{diag}\left(\frac{1}{N} \sum_{i=1}^N g_i \odot g_i\right)$, which better models gradient curvature but no longer admits an analytical solution; we instead perform a grid search over $w_{\text{SFT}} \in [0, 1]$ with 101 equally spaced grid points.

Table 2: GSM8K accuracy under zero-shot evaluation.

Method	Accuracy
Base Model	3.8e-3
SFT	0.4556
GAIF w/ rank-1 GN	0.4667
GAIF w/ diagonal empirical Fisher	0.4776

Results. Both GAIF variants outperform the SFT baseline, demonstrating the effectiveness of adaptive loss weighting across tasks beyond retrieval⁴. The diagonal empirical Fisher variant further improves over rank-1, suggesting that a more accurate Hessian approximation yields additional gains when computational budget permits. The rank-1 variant remains a practical default given its closed-form solution and lower memory cost.

6 Conclusion

In this paper, we presented the iterative GAIF algorithm, an adaptive and scalable framework for fine-tuning LLMs. While requiring a comparable level of hyperparameter optimization effort and training data as SFT, GAIF demonstrates consistent improvements over SFT across multiple tasks, both in terms of overall performance and output validity and instruction adherence. These gains are enabled by GAIF’s ability to automatically adapt to the underlying data distribution, evaluation metrics, and training dynamics. Consequently, the proposed approach allows practitioners to place greater emphasis on data curation and selection when fine-tuning language models for downstream applications.

⁴Due to the constrained token budget, zero-shot prompting setup, and strict output parsing requirements, this evaluation setting is more challenging than commonly reported configurations, resulting in lower absolute accuracy.

7 Limitations

Task and evaluation scope. Our primary task (movie recommendation) relies on proprietary query data and LLM judges, which limits external reproducibility. We partially mitigate this by including GSM8K as a public benchmark, and we deliberately use models of different families and scales across the two tasks (Mistral-7B and Qwen2.5-1.5B-Instruct) to cover more ground. Validation on a broader range of tasks is left to future work.

Implementation complexity. The adaptive weighting requires custom training code that computes and accumulates per-task gradients across micro-batches (gradient accumulation), incurring additional time and memory cost beyond standard fine-tuning. For practitioners, we recommend first adopting the inner-outer loop with joint SFT+DPO training under fixed task weights: we observe that this already yields strong results with minimal added complexity, and the adaptive weighting can be introduced subsequently if further gains are desired.

Optimizer mismatch. Our adaptive weighting rule is derived under an SGD update, whereas we train with AdamW—a mismatch shared by related gradient-based multi-task methods (Désidéri, 2012; Yu et al., 2020; Liu et al., 2021; Chen et al., 2018), which are likewise motivated under plain gradient descent yet applied with adaptive optimizers. In preliminary experiments we implemented an Adam-aware variant of the weighting rule, but it required reconstructing Adam’s internal preconditioning terms that the optimizer does not expose, which was substantially slower and more memory-intensive and yielded no improvement; we therefore retain the SGD-derived rule.

References

- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. 2016. [Concrete problems in ai safety](#). *Preprint*, arXiv:1606.06565.
- Thomas Carta, Clément Romac, Thomas Wolf, Sylvain Lamprier, Olivier Sigaud, and Pierre-Yves Oudeyer. 2023. Grounding large language models in interactive environments with online reinforcement learning. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. 2018. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. [Training verifiers to solve math word problems](#). *Preprint*, arXiv:2110.14168.
- Jean-Antoine Désidéri. 2012. [Multiple-gradient descent algorithm \(mgda\) for multiobjective optimization](#). *Comptes Rendus Mathématique*, 350(5-6):313–318.
- Chongming Gao, Ruijun Chen, Shuai Yuan, Kexin Huang, Yuanqing Yu, and Xiangnan He. 2025. [SPRec: Self-play to debias LLM-based recommendation](#). In *Proceedings of the ACM Web Conference 2025*, WWW ’25. ACM.
- Jiwoo Hong, Noah Lee, and James Thorne. 2024. [Orpo: Monolithic preference optimization without reference model](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 11170–11189. Association for Computational Linguistics.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Léo Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. [Mistral 7b](#). *Preprint*, arXiv:2310.06825.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. 2017. [Overcoming catastrophic forgetting in neural networks](#). *Proceedings of the National Academy of Sciences*, 114(13):3521–3526.
- Hanyu Lai, Xiao Liu, Junjie Gao, Jiale Cheng, Zehan Qi, Yifan Xu, Shuntian Yao, Dan Zhang, Jinhua Du, Zhenyu Hou, Xin Lv, Minlie Huang, Yuxiao Dong, and Jie Tang. 2025. [A survey of post-training scaling in large language models](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2771–2791. Association for Computational Linguistics.
- Bo Liu, Xingchao Liu, Xiaojie Jin, Peter Stone, and Qiang Liu. 2021. Conflict-averse gradient descent for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 34, pages 18878–18890.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder,

- Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Yu Meng, Mengzhou Xia, and Danqi Chen. 2024. [SimpO: Simple preference optimization with a reference-free reward](#). In *Advances in Neural Information Processing Systems*.
- Paul Michel, Sebastian Ruder, and Dani Yogatama. 2021. Balancing average and worst-case accuracy in multitask learning. *arXiv preprint arXiv:2110.05838*.
- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Qwen, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, and 24 others. 2024. [Qwen2.5 technical report](#). *Preprint*, arXiv:2412.15115.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. 2011. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the 14th International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and characterizing reward gaming. In *Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS '22*, Red Hook, NY, USA. Curran Associates Inc.
- Fahim Tajwar, Anikait Singh, Archit Sharma, Rafael Rafailov, Jeff Schneider, Tengyang Xie, Stefano Ermon, Chelsea Finn, and Aviral Kumar. 2024. Preference fine-tuning of LLMs should leverage suboptimal, on-policy data. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 47441–47474. PMLR.
- Alan D. Thompson. 2024. [Amazon nova \(formerly olympus\)](#). Parameter estimates based on benchmark analysis.
- Wei Xiong, Hanze Dong, Chenlu Ye, Ziqi Wang, Han Zhong, Heng Ji, Nan Jiang, and Tong Zhang. 2024. Iterative preference learning from human feedback: Bridging theory and practice for RLHF under KL-constraint. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 54715–54754. PMLR.
- Xuanhua Yang, Jianxin Zhao, Shaoguo Liu, Liang Wang, and Bo Zheng. 2023. Gradient coordination for quantifying and maximizing knowledge transfer in multi-task learning. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*.
- Xi Ye, Ruoxi Sun, Sercan O. Arik, and Tomas Pfister. 2024. Effective large language model adaptation for improved grounding and citation generation. In *Proceedings of NAACL*.
- Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. 2020. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Weizhe Yuan, Richard Yuanzhe Pang, Kyunghyun Cho, Xian Li, Sainbayar Sukhbaatar, Jing Xu, and Jason Weston. 2024. Self-rewarding language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 57905–57923. PMLR.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*.